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Department of Computer Engineering A.Y. 2025-26

PROJECT SYNOPSIS

ON

HEMOCONNECT – SMART BLOOD DONATION PLATFORM

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IN

COMPUTER ENGINEERING

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Project Guide

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HOD

1. Aim

The aim of the proposed project is to come up with an innovative system which is effective and smart called HemoConnect. This will connect the donor, patient, and hospital to minimize delay in providing blood in emergencies through geographic and prioritized matching.

2. Introduction

Blood donation is vital when it comes to rescuing people in various medical crises. Nevertheless, the traditional blood donation system is characterized by several limitations including inefficiencies, poor management of time, lack of real-time data, and inadequate coordination between the donor and the patient.

HemoConnect refers to a full-stack website that offers solutions to these issues by facilitating the management of the entire process of blood donation. Users have an opportunity to sign up as either donors or recipients, initiate blood requests, and monitor the progress of the request in real time. Hospitals act as central institutions in validating the authenticity of the request and the donor.

3. Literature Survey

An analysis of existing systems for blood donation shows that there are several different ways through which donor management and availability can be improved. While traditional blood bank systems are mainly based on recording information about donors and controlling blood stock, there is usually an absence of real-time communication and cooperation between donors and recipients. Some recent studies, conducted by S. Sulaiman et al. and S. Ka, A. Rai, have provided information regarding simple and complex management systems that allow collecting donor information and processing requests. The geofencing technique applied by R. Dandekar et al. makes use of location-based services in order to find nearby donors and thus improve efficiency. However, even centralized systems such as e-RaktKosh that control data management on a national level still lack sufficient features of real-time donor management and users' interaction. It becomes clear from the analysis of existing methods that donor management is currently supported through various techniques; however, all the techniques are not combined in one system.

4. Motivation and Challenges

Motivation

The motivation for developing an effective blood donation system arises from the frequent issues of delays and complications experienced in emergency situations. Due to lack of information and inefficient coordination between patients, donors, and hospitals, it becomes very difficult for some patients to get compatible donors who may save their lives.

The rise of modern internet technologies, including smartphones, provides a good opportunity for designing a platform that would allow all stakeholders to coordinate their efforts. The motivation of the HemoConnect system is to create an automated platform using modern web development technologies that would facilitate communication between patients, donors, and hospitals.

Challenges

- Ensuring security of medical information.
- Providing real-time updates and avoiding high latency
- Ensuring accurate matching based on blood groups and locations
- Handling many users at the same time
- Integrating external APIs such as OpenStreetMap
- Providing reliable service
- Encouraging user participation

5. Objectives and Scope of the Project

Objectives

- Creating a centralized web-based platform for blood donations
- Facilitating real-time matching of donors and recipients based on location
- Implementing verification from hospitals in order to ensure reliability of the service
- Reducing response time
- Ensuring secure data management and processing

Scope of the Project

The scope of the current project includes creation of the web application for donating blood, in which the user can log in and place his/her requests. The users of the application will be divided into three categories: the donor, the patient who needs blood, and the hospital where the patient is staying. The patient can request for blood donation, and then the donor accepts the invitation and arranges the meeting with the patient.

Other critical functionalities of the system include the health testing of the donor. There might be some possibilities to develop the system further to add more features such as the creation of a mobile app and integration of AI-driven algorithms. Currently, the system is developed to handle problems with blood donations.

6. Proposed Methodology

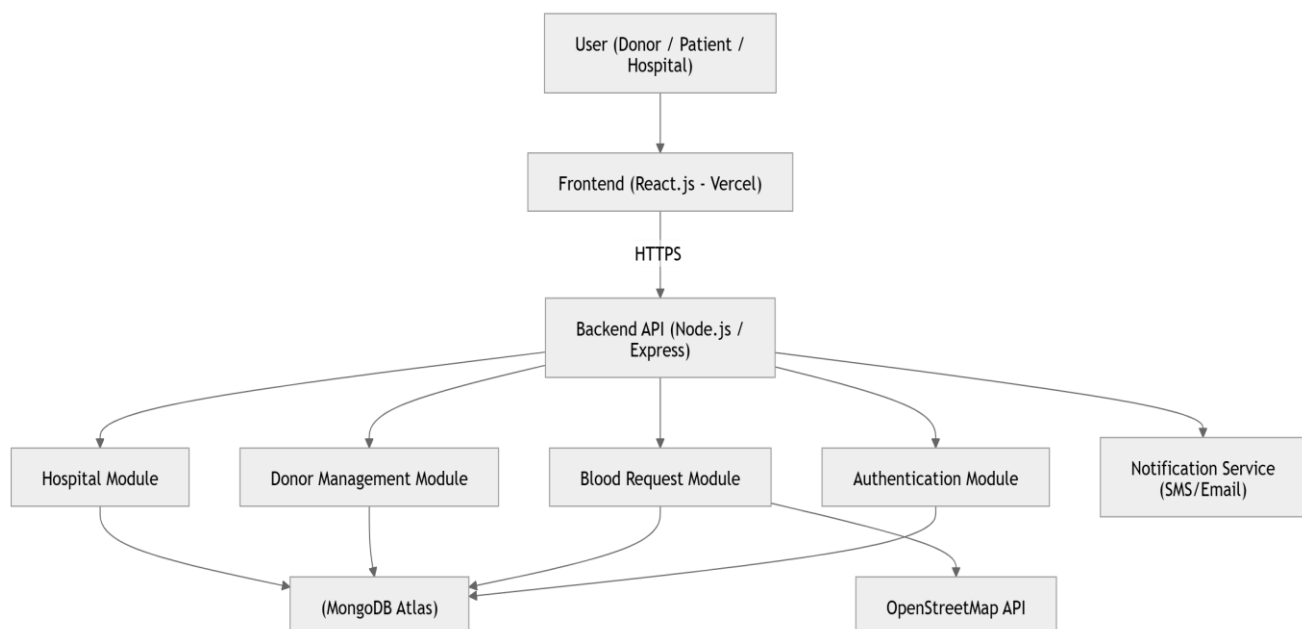


Figure 1: System Architecture

The system follows a modular architecture:

System Architecture:

- Frontend: React.js (Vercel)
- Backend: Node.js with Express
- Database: MongoDB Atlas

- APIs: OpenStreetMap, Notification APIs

Modules:

- Authentication Module
- Blood Request Module
- Donor Management Module
- Hospital Verification Module
- Notification System

Workflow:

- User registers/logs in
- Creates blood request
- Hospital verifies request
- System matches donor
- Donation scheduled
- Notification sent

7. Software And Hardware Requirements

Software requirement

Sr. No.	Component	Specifications
1	Operating System	Windows/Linux
2	Technology	React.js, Node.js, Express
3	Tool	VS Code, GitHub, Postman
4	Database	MongoDB
5	Server	Vercel

Hardware requirement

Sr. No.	Component	Specifications
1	Processor	Intel i5 or above
2	Memory	Minimum 8GB RAM
3	Network Interface	Internet connection
4	Other (If required)	Smartphone/Laptop

8. Contribution of the project to the field of Technology and Society

Technological Contribution:

- Implementation of a full-stack web application using modern technologies (React, Node.js, MongoDB)
- Integration of location-based services using OpenStreetMap
- Development of a modular and scalable system architecture
- Use of real-time data handling and API-based communication

Societal Contribution:

- Helps in saving lives by reducing delays in blood availability
- Promotes awareness and participation in blood donation
- Improves coordination between donors, patients, and hospitals
- Provides a transparent and reliable system for healthcare services

9 Limitations

- The system requires a stable internet connection to function
- Currently limited to a web-based platform (no mobile application)
- Real-time updates are based on polling instead of WebSockets
- Accuracy depends on user-provided information
- Limited integration with external healthcare systems
- AI-based prediction and automation features are not yet implemented

10 Project Plan:

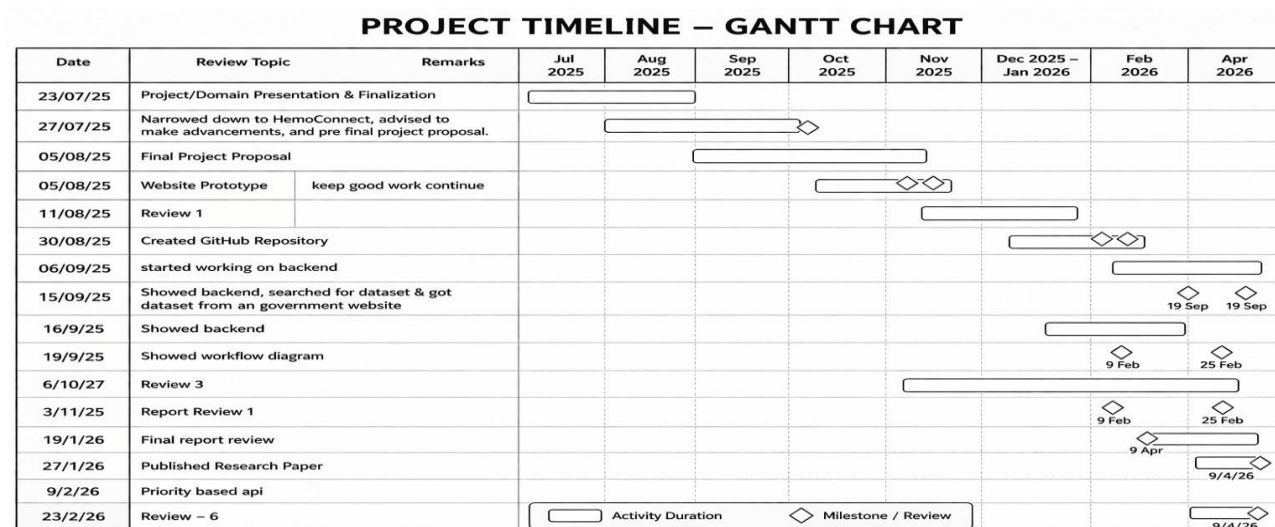


Figure 2: Project Timeline-Gantt Chart

11 Conclusion

This project is called HemoConnect and is the most efficient and advanced solution to blood management that will address the deficiencies typical of existing systems. Thanks to integration of the latest achievements in the development of web technology, the project provides fast communications, safety of confidential data, and proper cooperation between all participants of the process.

By implementing such an innovative project, we demonstrate that full-stack technologies can be efficiently utilized in dealing with healthcare problems. The project proves to be useful since it reduces response time, increases users' satisfaction, and assures proper functioning of the application. After some necessary enhancements, the application will have a high prospect of being applied in the healthcare field.

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